NyaySetu

Product Requirements Document

Making every citizen–police interaction visible, verifiable and continuous

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1. Problem Statement

A citizen can walk into a police station, ask for help, explain an incident, receive some form of acknowledgement, and walk out — without knowing whether their interaction was formally recognised, what was officially recorded, whether that record reflects what they said, what happened after they left, whether they need to do anything next, or whether returning to follow up will mean starting their story over.

That is the gap NyaySetu exists to close.

The problem is not that citizens lack a tracking page. The deeper problem is structural:

> **A citizen can enter an institutional workflow and then become invisible to themselves inside it.**

This invisibility is not evenly distributed. It falls hardest on people who arrive without confidence, legal knowledge, institutional familiarity, physical proximity, or the ability to follow up repeatedly. As uncertainty and effort rise, these citizens are the first to disengage from an interaction that was never resolved.

NyaySetu is designed around three questions the citizen is actually asking:

- **Entry** — "I came here for help. Can the system acknowledge that?"
- **Fidelity** — "Is this what was officially recorded?"
- **Continuity** — "What is happening now, and what should I do next?"

The product challenge follows directly:

> *How might we make a citizen's interaction with the police traceable, verifiable and continuous — without replacing police operational systems, and without creating a burden-heavy second workflow for station staff?*

2. Project Objective

NyaySetu is a **citizen continuity layer** for high-stakes public interactions, beginning with policing.

It is not a police operating system. It does not investigate crime, determine legal classifications, manage investigations, or replace official records. It sits around the official interaction and does one thing well: it keeps the citizen visible to themselves from the moment they seek help through to case follow-up.

Concretely, the objective is to deliver a two-sided system that lets a citizen:

- create a traceable **Visit Reference** the moment they seek help;
- describe the incident in their own words, before the interaction begins;
- receive a clear acknowledgement that their interaction took place;
- review a plain-language representation of what was officially recorded;
- confirm that representation, or flag a discrepancy — without editing the official record;
- understand the current state of their interaction and their **one clear next action**;
- add newly remembered information against the same reference;
- and return for a follow-up without repeating their entire story.

And, on the other side, lets **authorised station staff** create those citizen-visible updates quickly, through structured actions, without maintaining a second full case-management system.

The guiding promise:

> **Never leave a citizen wondering what happens next.**

Or, in the citizen's own emotional terms:

You shouldn't need power, legal knowledge, or repeated visits just to know that you were heard.

3. Scope of the Project

NyaySetu's MVP covers **three connected layers** of a single citizen journey, delivered across two product surfaces.

In scope — the three layers

- **Entry.** The citizen selects a station, states a reason for the visit, describes the incident in free text, optionally attaches supporting information, and receives a unique Visit Reference — created before the physical interaction, so their attempt to seek help is traceable from the start.
- **Fidelity.** After the interaction is recorded, the citizen receives a readable representation of what was recorded, and can either confirm it or flag that something is missing or incorrect. Their disagreement becomes traceable; it never silently rewrites the official record.
- **Continuity.** After leaving, the citizen has one place to see current status, a meaningful timeline, the latest update, and exactly one next action — with the ability to add information or generate a follow-up reference that carries context forward.

In scope — the two surfaces

- The **Citizen Experience**, a mobile-first application through which the citizen enters, remains visible, reviews, and continues the journey.
- The **Station Console**, a lightweight station-side surface through which authorised staff locate an interaction, review submitted context, and publish structured, citizen-safe state changes.

The **primary end-to-end scenario** the MVP must demonstrate is a single high-friction journey: lost property following a problematic cab interaction. One station, one journey, validated first.

Anything beyond making entry, representation and continuity visible for that journey is out of scope for the MVP; the explicit exclusions are enumerated in Section 14.

4. Target Users & Stakeholders

4.1 Primary user — the everyday citizen seeking help

The primary user is deliberately not assumed to be legally knowledgeable, confident with authority, digitally sophisticated, familiar with police procedure, or able to travel repeatedly to a station. They may be stressed, tired, confused, elderly, alone, and unsure which details even matter.

> **Primary Job to Be Done**
> *When I approach the police for help after an incident, help me enter the interaction clearly,*

> *understand what was recorded, and know what happens next — so that I do not need influence, legal knowledge, or repeated visits just to stay connected to my own interaction.*

4.2 Secondary user — station staff

The Station Console supports authorised station personnel who interact with citizens or maintain the station-side continuity workflow. The product must not assume that one officer owns an interaction forever, that every update is hand-written by an investigating officer, or that the station will maintain a second full case-management system. The station acknowledges the visit; an officer may become the current interaction owner; the citizen's experience must not depend on any one officer permanently owning the case.

> **Station Staff Job to Be Done**
>
> *When a citizen interaction needs acknowledgement or a meaningful citizen-facing update, help me publish the right state quickly — without duplicating operational work or exposing internal information.*

4.3 Stakeholders

Stakeholder	Primary need
Citizen	Clarity, acknowledgement, continuity
Station staff	Fast context retrieval and low-burden updates
Frontline officer	Reduced repeated storytelling
Station leadership	Visibility into interaction bottlenecks
Operations	Workflow analytics
Policy / oversight	Aggregate interaction insights
Design	Accessible, low-stress experience
Engineering	Reliable state management and integrations
QA	Correct transitions and traceability
Data	Privacy-conscious workflow analytics

Two principles hold this table together:

> **NyaySetu must not optimise citizen convenience by creating an unusable station workflow.**
>
> **NyaySetu must not optimise internal efficiency by making citizens invisible.**

The product lives at the interface between the two, and must not sacrifice one for the other.

5. Evidence & Research Framing

Because of the case-study timeline, primary interviews were not conducted. The opportunity is therefore developed as an **evidence-backed hypothesis**, drawing on published public-perception research, a detailed real-world citizen journey, and secondary ecosystem research. Key assumptions are labelled as such (Section 16) and should be validated before any production rollout. Nothing here implies that primary user interviews occurred.

5.1 What the public-perception evidence shows

The research base points consistently at a trust-and-visibility gap between citizens and police — and, importantly, at the two specific failure points NyaySetu targets.

A long tradition of scholarship describes an avoidance syndrome in the Indian public's relationship with the police — a substantial unwillingness to approach or engage, more pronounced among poorer, less-literate and marginalised citizens, and worsened by prior negative contact. The public avoids contact with the police, and this avoidance is more pronounced in rural than urban areas, with a significant section of the public not feeling at ease dealing with the police [Ref. 1]. Critically for this product, the same assessments identify two concrete breakdowns:

- **The continuity gap.** Complainants report that they do not get information regarding the status of their complaints, and that such information is furnished only after repeated visits to the police station [Ref. 1]. This is exactly the "what happens after I leave?" failure NyaySetu's Continuity layer addresses.
- **The fidelity gap.** There is often a difference between the complainant's version and the description of the case as recorded in the file by the police [Ref. 1]. This is precisely the "is this what was actually recorded?" failure NyaySetu's Fidelity layer addresses.

Structural analyses locate the roots of this distance in a colonial-era institutional design oriented toward control rather than citizen service, compounded by understaffing, weak accountability and limited public trust — conditions that persist despite decades of reform calls [Ref. 3]. The most recent large-scale evidence reinforces that the citizen-facing experience cannot be assumed to be neutral or reliable: the Status of Policing in India Report 2025, based on a survey of 8,276 police personnel across 17 states and a Union Territory, documents widespread disregard for due process and accountability from within the force itself [Ref. 2, Ref. 4].

The pattern across these sources is consistent: citizens are uncertain whether they will be heard, uncertain whether they were recorded faithfully, and uncertain what is happening after they leave — and that uncertainty falls hardest on those with the least power to push back.

5.2 The citizen journey that made the problem concrete

A real incident shared during discovery grounds the abstraction.

A citizen lost a purse after being unexpectedly asked to leave a private cab. In the confusion, the vehicle registration number was never captured. The citizen approached a police station for help, and the experience broke down across a series of moments: uncertainty over whether the incident would be taken seriously; reluctance perceived in the initial interaction; dependence on a more confident, more legally-aware family member to steer the conversation; unclear acknowledgement at the end; no clear way to continue remotely; and no sense of whether anything happened after they left.

Eventually, the citizen and family disengaged. Months later, they still did not know what had happened, whether any action had continued, whether they needed to do anything, or how to re-engage.

The purse was the immediate loss. The more important product failure was this:

> **The citizen lost connection with the system.**

5.3 Evidence hierarchy

The case study distinguishes clearly between what is observed and what is hypothesised:

Type	Framing
Public-perception research (Sec. 5.1)	Observed pattern
Real cab/purse incident (Sec. 5.2)	Detailed journey evidence
Secondary ecosystem research	Secondary evidence
The product opportunity	Hypothesis
The MVP	Intervention to test the hypothesis

6. Product Definition & Architecture

6.1 What NyaySetu is

NyaySetu is a **two-sided citizen continuity system**:

- **A. Citizen Experience Layer** — through which a citizen enters an interaction, remains visible, reviews information, understands updates, and continues the journey.
- **B. Station Continuity Layer** — through which authorised staff acknowledge meaningful interactions, surface selected information for review, publish structured citizen-safe updates, receive traceable feedback, and communicate next actions.

Together they form a single shared interaction trail — the bridge between institutional workflow and citizen understanding.

6.2 What NyaySetu is not

NyaySetu does not investigate crime, identify suspects, recover property, determine legal classifications, manage investigations, or replace the official record. It adds a citizen-visible layer around the interaction; it never becomes the police operating system. (Full exclusions: Section 14.)

6.3 The central distinction: System of Record vs. System of Citizen Experience

This distinction is the architectural spine of the product, and it must never blur.

Existing police systems OWN	NyaySetu OWNS
Official operational records	Citizen interaction references
Investigation workflows	Citizen-facing continuity states
Internal classifications	Acknowledgement timestamps

Existing police systems OWN	NyaySetu OWNS
Institutional decision-making	Record-review events
Operational details	Discrepancy feedback references
—	Citizen-submitted supplementary information
—	Next-action communication

NyaySetu does not replace the official record. It creates a citizen-visible layer around the interaction — and it must be **architecturally incapable** of publishing arbitrary internal operational text to citizens. Only authorised structured states (Section 7) can ever reach the Citizen Experience.

6.4 Why the Citizen Experience alone is insufficient

A citizen-facing app cannot manufacture truthful continuity on its own. For the citizen to receive meaningful updates, something must create those updates from a trustworthy source. This is the reason the product is two-sided rather than one:

> **Continuity requires an interaction source of truth and a controlled mechanism for publishing citizen-safe state changes — and that is what the Station Console provides.**

6.5 MVP architecture

During the MVP, the Station Console is the controlled source of truth for citizen-safe interaction events:

```
Station Staff
   ↓
NyaySetu Station Console
   ↓
Structured Interaction State
   ↓
NyaySetu Continuity Layer   (append-only interaction event store)
   ↓
Citizen Interaction Home
   ↓
Citizen
```

The Continuity Layer is built behind a **clean ingress boundary** — an internal “interaction-events” API that does not care whether an event came from a human clicking the console or, later, from an integration. This is what lets production swap the source without a rewrite:

```
Existing Police Operational Systems
   ↓
Approved workflow events / interoperability APIs
   ↓
NyaySetu Continuity Layer
   ↓
Citizen-safe state translation
```

```
   ↓
Citizen Experience
```

The MVP validates the interaction model first. The production integration question comes later — and only if the model earns it.

7. The Structured Update Model

This is the heart of NyaySetu. Everything else is scaffolding around it.

Institutions think in **process states**. Citizens experience **uncertainty states**. An internal system says "Pending"; the citizen thinks "Should I do something?" An internal system says "Under process"; the citizen thinks "Is anybody actually doing anything?" The station may hold a perfectly valid internal record while the citizen still feels "I don't even know if I entered the system properly."

NyaySetu's job is to translate between those two worlds. Every citizen-facing change flows through one pipeline:

```
Operational action  →  Structured interaction state  →  Citizen meaning  →
Citizen next action
```

Staff **never write free-form updates to citizens** in the MVP. Citizen-facing copy is derived from the state, not typed by a human. This is enforced in code, not left to discipline — it is what makes the system architecturally incapable of leaking internal operational text (Section 6.3), and it is what guarantees every update carries citizen meaning rather than system jargon.

The MVP supports exactly this set of structured states:

Station-side event	Citizen-facing meaning	Citizen next action
Visit created	Your visit reference has been created	Attend / continue interaction
Interaction acknowledged	Your interaction has been acknowledged	Wait for the next update
Recorded information available	Your recorded information is ready to review	Review information
Citizen confirms record	Your confirmation has been recorded	No action unless updated
Citizen flags discrepancy	Your feedback has been received	Wait for acknowledgement / next action
More information required	We need something from you	Submit information
Follow-up required	Please visit the station again	Generate / use follow-up reference
No action required	You do not need to do anything right now	No action

The right-hand column is deliberately constrained. **Every next action resolves to exactly one of three:** No action required · Information required · Visit required. At any moment, the citizen knows which of the three they are in — and never has to guess.

7.1 Language guardrails (enforced in code and tests)

The translation is not just for clarity; it is an ethical boundary. These rules are tested, not merely documented:

- Acknowledgement copy must **not** imply case registration or investigation acceptance.
- The citizen is **never** shown "investigation in progress" merely because a Visit Reference exists.
- Discrepancy copy is "Your feedback has been received." It must never say "The record has been corrected" or "The station agrees the record was wrong."
- Record confirmation means "This reflects what I understand was recorded" — never legal validity or a promise about outcome.
- Timeline events must carry citizen meaning. Generic noise such as "Status updated" is forbidden.

7.2 Discrepancy handling

Discrepancy handling is where the Fidelity pillar earns its keep, and it depends on one hard rule: **citizen feedback and the official record are always distinct, and feedback never overwrites the record.**

When a citizen flags a discrepancy:

1. NyaySetu records the citizen's feedback as a separate, traceable concern.
2. The official record remains unchanged.
3. Authorised staff can see that a discrepancy was raised.
4. Staff can acknowledge the concern.
5. Resolution, if any, happens through the relevant official workflow — outside NyaySetu.

The system may say "Your feedback about the recorded information has been received." It may never imply the record was changed or that the station conceded the point.

This is one of NyaySetu's two genuine differentiators. Ordinary tracking systems answer "where is my case?" NyaySetu also answers a question they skip entirely: **"did the system accurately represent what I said?"** The other differentiator is the translation layer itself — turning institutional state into citizen meaning and a single next action.

8. Modules & Major Features

The MVP is organised into nine feature groups — seven on the Citizen Experience, two on the Station Console. Design may consolidate the citizen-facing moments into fewer screens (Section 12), but the functional groups below are the build units.

Citizen Experience

A. Start a Visit. The citizen selects or confirms a station, chooses a broad reason for the visit (Lost property, Theft, Harassment, Safety concern, Other), describes the incident in free text with no legal terminology required, optionally attaches supporting information, and receives a unique **Visit Reference**. The reference is clearly labelled as not an official complaint or registered case.

B. Interaction Status. The citizen tracks a small, honest set of states — Visit Created → Awaiting Interaction → Interaction Acknowledged. Status advances only on a valid station-side event, and acknowledgement timestamps are stored.

C. Record Verification. The citizen reads a plain-language representation of what was officially recorded, then either confirms it reflects what they reported or flags a discrepancy. A flag creates a separate, traceable concern with its own reference; the citizen's feedback stays visibly distinct from the record.

D. Interaction Home. The primary citizen hub — internally named "Interaction Home," never assumed to be a "case." It shows the interaction title, current citizen-facing state, latest meaningful update, next action, and access to both the timeline and the recorded information.

E. Interaction Timeline. A sequence of meaningful milestones only — Visit Reference created → interaction acknowledged → recorded information available → record reviewed → confirmed or discrepancy flagged → current state. Meaningless events are banned.

F. Add New Information. The citizen submits newly remembered information against the existing reference — free text plus optional attachment, timestamped, associated to the interaction, and explicitly confirmed on submission — without restarting the interaction.

G. Follow-Up Visit. The citizen generates a follow-up reference linked to the parent interaction, choosing a reason (asked to return / new information / need clarification / other). Prior context travels with the follow-up, so the citizen never explains everything from the beginning again.

Station Console

The Station Console exists for one purpose: **to reliably create meaningful citizen-visible state changes without requiring staff to maintain a second full operational system.** It is two screens.

H. Station Inbox. Authorised staff search by Visit Reference, locate interactions needing attention, and see a high-level reason, the current citizen-facing state, and an attention indicator. It must not become an investigation dashboard, and it must not expose sensitive detail in list views. Staff never recreate citizen-provided context — it is already there.

I. Interaction Detail. Staff review the full citizen context — Visit Reference, reason, submitted description, attachment indicators, timeline, current state, and any outstanding discrepancy feedback — then publish the next state through a constrained set of **structured actions**: Acknowledge Interaction · Make Recorded Information Available · Request Additional Information · Publish No Action Required · Request Follow-Up. Every action creates a traceable event, maps to an authorised state, and updates the Citizen Experience.

The golden end-to-end journey

The MVP must demonstrate the whole loop, not isolated screens. Primary scenario: lost property following a problematic cab interaction.

```
Seek help → Become visible → Interaction acknowledged → Recorded information
available
→ Citizen verifies representation → Citizen remains connected → Meaningful update
→ Clear next action
```

9. Functional Requirements

Requirements are grouped by feature. Each carries acceptance criteria that a build can be tested against.

FR-1 · Start a Visit (Feature A)

The system shall let a citizen start a visit, select or confirm a police station, select a broad reason, describe the incident in free text, optionally attach supporting information, and generate a unique Visit Reference.

Acceptance criteria:

- A unique reference is generated and linked to the selected station.
- The citizen can view exactly what they submitted.
- Optional attachments never block progression.
- Free-text input never requires legal terminology.
- The system visibly distinguishes a Visit Reference from an officially registered complaint or case.

FR-2 · Interaction Status (Feature B)

The system shall show the citizen whether their interaction has been acknowledged, across the states Visit Created → Awaiting Interaction → Interaction Acknowledged.

Acceptance criteria:

- Status advances only when a valid station-side event occurs.
- A citizen is never shown “investigation in progress” solely because a Visit Reference exists.
- Acknowledgement timestamps are stored.
- Acknowledgement language never implies case registration or investigation acceptance.

FR-3 · Record Verification (Feature C)

The system shall let a citizen review a readable representation of what was recorded, then confirm it or flag a discrepancy.

Acceptance criteria:

- Record content is readable and understandable without procedural knowledge.
- Confirmation is timestamped.
- A discrepancy creates a separate, traceable concern and returns a discrepancy reference.
- Flagging never silently overwrites the official record.
- Citizen feedback remains distinguishable from the official record.

FR-4 · Interaction Home (Feature D)

The system shall provide a single hub showing interaction title, current citizen-facing state, latest meaningful update, next action, and access to the timeline and recorded information.

Acceptance criteria:

- The citizen can always resolve their situation to exactly one of: No action required, Information required, Visit required.

FR-5 · Interaction Timeline (Feature E)

The system shall display meaningful milestones for the interaction.

Acceptance criteria:

- Every timeline event communicates citizen meaning.
- Non-meaningful events (e.g. “Status updated”) are not emitted.

FR-6 · Add New Information (Feature F)

The system shall let a citizen add information to an existing interaction via free text and optional attachment.

Acceptance criteria:

- Submissions are timestamped and associated to the existing interaction.
- The citizen receives explicit submission confirmation.

FR-7 · Follow-Up Visit (Feature G)

The system shall let a citizen generate a follow-up reference linked to the parent interaction, with a stated reason.

Acceptance criteria:

- The follow-up reference links to the existing interaction.
- Relevant prior context is available during the returning workflow.
- The citizen can see why the follow-up reference was created.

FR-8 · Station Inbox (Feature H)

The system shall let authorised staff search by Visit Reference and locate interactions needing attention.

Acceptance criteria:

- Search by Visit Reference returns the relevant interaction.
- Staff never need to recreate citizen-provided context.
- Sensitive details are not exposed in list views.
- The inbox does not function as an investigation dashboard.

FR-9 · Interaction Detail & Structured Actions (Feature I)

The system shall let authorised staff review a citizen interaction and publish a meaningful next state via structured actions.

Acceptance criteria:

- Every staff action creates a traceable event.
- Every citizen-facing update maps to an authorised state.
- The system prevents publication of arbitrary internal operational information.
- Actions never imply investigation or case registration unless explicitly supported by the source system.
- Published states update the Citizen Experience.

FR-10 · Structured Update Enforcement (cross-cutting)

The system shall derive all citizen-facing copy from authorised structured states (Section 7); staff shall not be able to enter free-form citizen-facing prose in the MVP.

Acceptance criteria:

- No code path allows arbitrary staff text to reach the citizen.
- Every state maps to a defined citizen meaning and a next action in {No action required, Information required, Visit required}.

10. Non-Functional Requirements

NyaySetu operates in a high-stakes public setting, for users who are often stressed and non-expert. The non-functional bar reflects that: visibility must never be bought at the cost of privacy, safety, or dignity.

NFR-1 · Privacy

Data minimisation, purpose limitation, and appropriate retention controls throughout. No personally identifying information in logs or analytics. Analytics use privacy-conscious identifiers only.

NFR-2 · Security

Encrypted transport and encryption at rest; role-based access control for all station staff; full audit logging of staff actions; secure attachment handling (type and size limits, scanning, signed URLs, no public buckets).

NFR-3 · Reliability

No silent loss of submissions. Every submit returns explicit confirmation. Behaviour on connectivity loss is graceful and explicit, never ambiguous. Writes to the interaction event store are idempotent.

NFR-4 · Accessibility (WCAG AA)

A first-class requirement, not a polish pass. The experience is designed for older adults, users with low digital confidence, users under stress, and users of assistive technology — large tap targets, plain language, high contrast, screen-reader labels, and minimal cognitive load.

NFR-5 · Performance

Core actions — Visit Reference generation, interaction lookup, acknowledgement publishing — feel immediate and never introduce additional waiting.

NFR-6 · Maintainability & Extensibility

The Continuity Layer sits behind a clean ingress boundary (Section 6.5), so the MVP's console-as-source can later be swapped for approved workflow events or interoperability APIs without a schema rewrite. Citizen auth can upgrade from reference-based access to real identity without redesign.

NFR-7 · Prototype vs. Production honesty

Wherever a production dependency is simulated in the MVP (auth, notifications, record source, attachment scanning), the prototype must clearly distinguish **prototype behaviour** from **production dependency** — in the interface and in the code.

11. External Interfaces & the Citizen ↔ Station State Contract

11.1 Interfaces

The two surfaces behave as one connected system, joined by a single internal interface and a set of simulated external ones.

- **Internal interaction-events API** — the ingress boundary to the Continuity Layer. The Station Console (and, in production, approved integrations) publishes structured events here; the Citizen Experience reads projections of them. This is the only channel by which anything reaches the citizen.
- **Attachment storage interface** — object storage with signed URLs; simulated in the MVP but abstracted behind a stable interface.
- **Notification interface** — abstracted behind an interface; the MVP may use a clearly-labelled stub. No production notification dependency is assumed.
- **Identity / auth interface** — RBAC for staff; lightweight reference-based access for citizens, designed to upgrade to real identity later.

11.2 The state contract

The two surfaces must never disagree about the state of an interaction. This matrix **is the contract** — a state transition not represented here is treated as impossible, and it is written up as automated integration tests (Section 13.3 shows the same truth as a state machine).

Citizen event	Station visibility	Station action	Citizen outcome
Visit created	New interaction available	Review / acknowledge	Awaiting interaction
Interaction attended	Interaction visible	Acknowledge	Interaction acknowledged
Record available	Detail updated	Publish record-ready	Review information
Discrepancy submitted	Feedback visible	Acknowledge feedback	Feedback acknowledged
More info needed	Requirement identified	Request information	Information required
Follow-up needed	Requirement identified	Request follow-up	Visit required

> *Follow-up modelling note. "Follow-up needed" resolves the citizen outcome to Visit required, but it does not reopen the current interaction. Requesting a follow-up spawns a new, linked interaction (its own lifecycle, from Visit Created) that carries prior context forward via parent_interaction_id — the original interaction's resolved state is preserved, never overwritten. See the state diagram (13.3) and data model (13.4).*

12. Use Cases / Scenarios

The MVP is validated against one end-to-end journey and the discrete use cases that compose it. The primary scenario throughout is lost property following a problematic cab interaction.

Actors: the **Citizen** (primary user), **Station Staff** (authorised console user), and the **System** (the Continuity Layer that stores events and derives citizen-safe state).

UC-01 · Create a Visit

Actor: Citizen · **Goal:** Make the attempt to seek help traceable before interacting.

Preconditions: Citizen has the app; a station is selectable.

Main flow: Citizen selects a station → chooses a reason → describes the incident in their own words → optionally attaches information → the system generates a Visit Reference.

Postcondition: A traceable Visit Reference exists, clearly marked as not an official complaint.

UC-02 · Acknowledge the Interaction

Actor: Station Staff · **Goal:** Confirm the interaction took place.

Preconditions: A Visit Reference exists; citizen has attended.

Main flow: Staff locate the interaction in the Inbox → open Interaction Detail → review submitted context → publish Acknowledge Interaction.

Postcondition: Citizen state advances to Interaction acknowledged, with a stored timestamp; no investigation is implied.

UC-03 · Make the Record Available & Verify Fidelity

Actor: Station Staff, then Citizen · **Goal:** Let the citizen check what was recorded.

Main flow: Staff publish Make Recorded Information Available → citizen opens a plain-language representation → citizen either confirms it reflects what they reported, or flags a discrepancy.

Alternate flow (discrepancy): A separate traceable concern is created with its own reference; the official record is untouched; staff can later acknowledge the feedback.

Postcondition: Either a timestamped confirmation, or a traceable discrepancy — the citizen's feedback distinct from the record throughout.

UC-04 · Stay Connected via Interaction Home

Actor: Citizen · **Goal:** Understand state without an unnecessary visit.

Main flow: Citizen opens Interaction Home → sees current state, latest meaningful update, timeline, and exactly one next action from {No action required, Information required, Visit required}.

Postcondition: The citizen knows what happened, what is happening, and what to do next.

UC-05 · Add New Information

Actor: Citizen · **Goal:** Supply newly remembered detail without restarting.

Main flow: Citizen opens the existing interaction → adds free text and optional attachment → receives explicit confirmation.

Postcondition: Timestamped information is associated to the interaction.

UC-06 · Generate a Follow-Up Visit

Actor: Citizen, then Station Staff · **Goal:** Return without repeating the whole story.

Main flow: Citizen selects a reason for returning → the system generates a follow-up reference that spawns a new interaction linked to the parent → prior context is available to staff on the returning workflow.

Postcondition: A new, linked interaction exists (via parent_interaction_id); the citizen does not re-explain from zero, and the original interaction's resolved state is preserved.

13. UML Diagrams

Four views of the system: who uses it, how the golden journey flows across surfaces, how interaction state transitions, and how the data is structured. All four are consistent with the state contract in Section 11.2.

13.1 Use Case Diagram

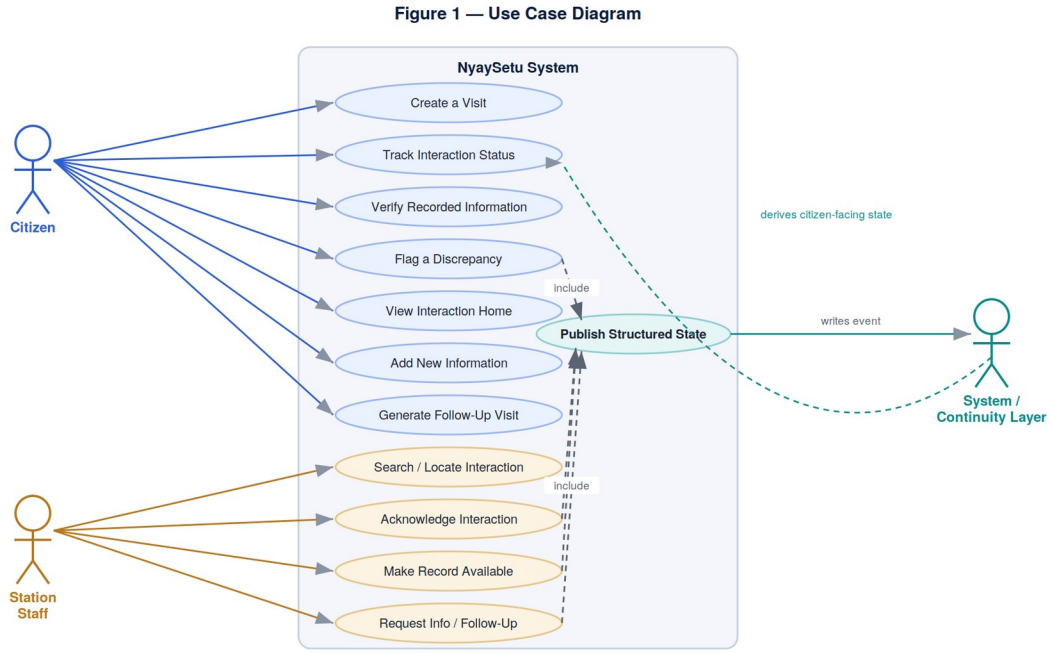


Figure 1 — Use Case Diagram

The citizen never publishes state directly; every citizen-visible change is a structured state published through the System. Flagging a discrepancy includes structured-state publication — creating a traceable concern rather than editing a record. Generating a follow-up spawns a new, linked interaction with its own lifecycle, not a re-entry of the current one.

13.2 Sequence Diagram — the Golden Journey

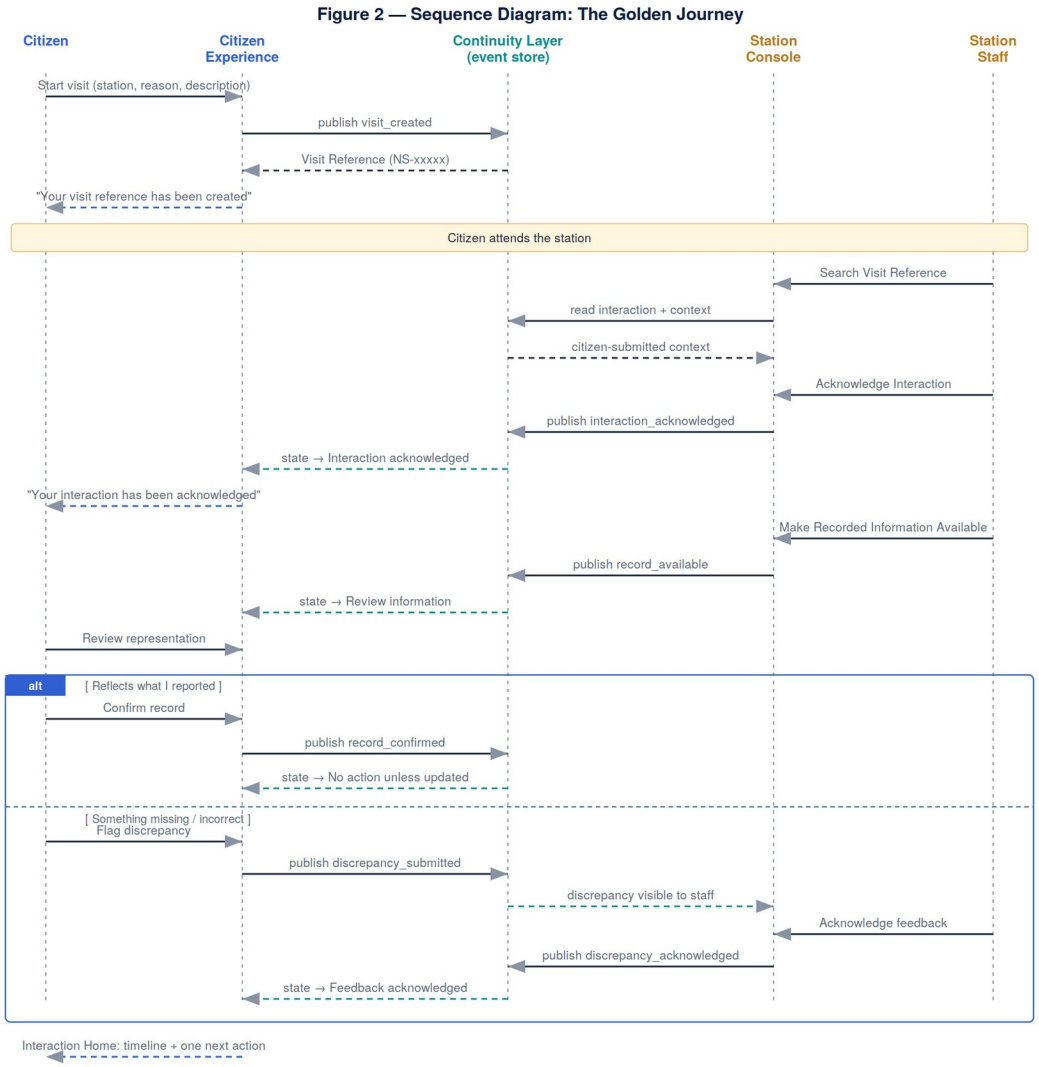


Figure 2 — Sequence Diagram: The Golden Journey

13.3 State Diagram — Interaction Lifecycle

Figure 3 — State Diagram: Interaction Lifecycle

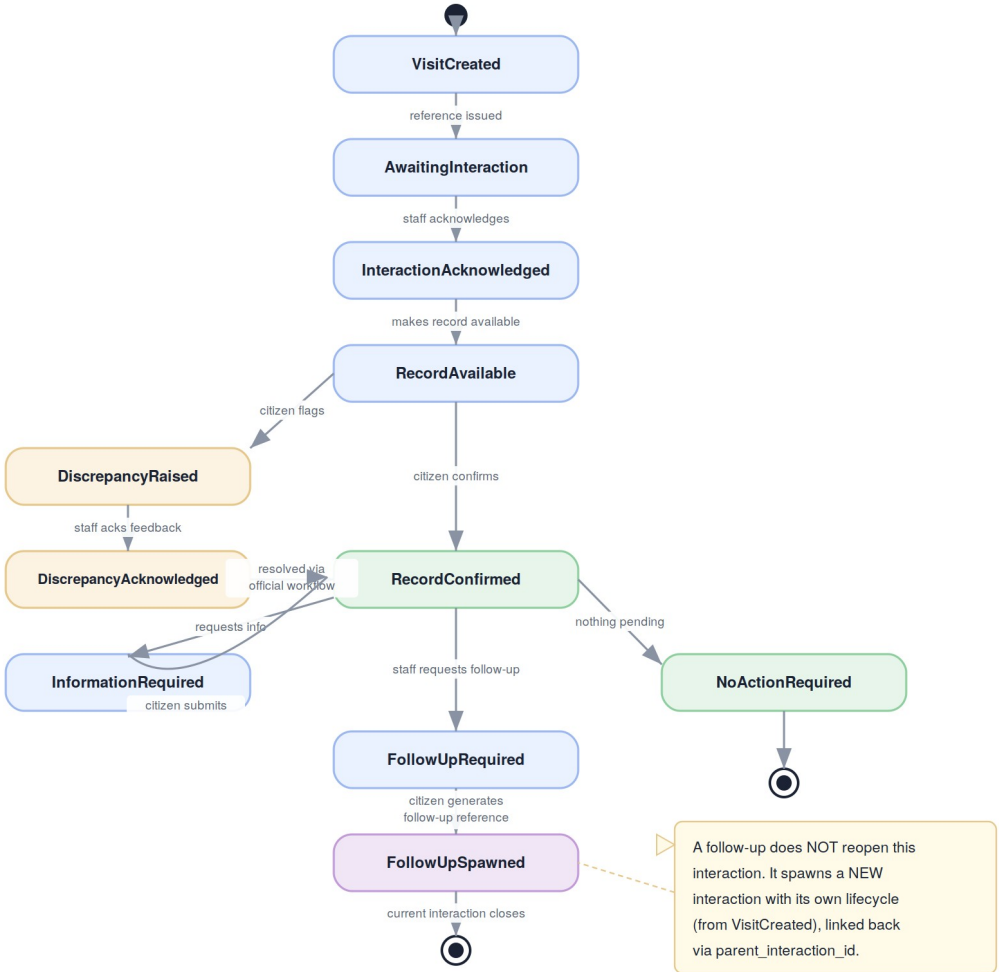


Figure 3 — State Diagram: Interaction Lifecycle

Every state maps to exactly one citizen next action in {No action required, Information required, Visit required}. A transition not shown here is not permitted — this diagram and the Section 11.2 contract are the same rule, enforced as integration tests. DiscrepancyRaised never mutates the record; it is a parallel, traceable concern resolved through the official workflow. A follow-up terminates the current interaction's active cycle and begins a new linked interaction that carries prior context forward — the original interaction's resolved state is preserved, never overwritten.

13.4 Class / Data Model Diagram

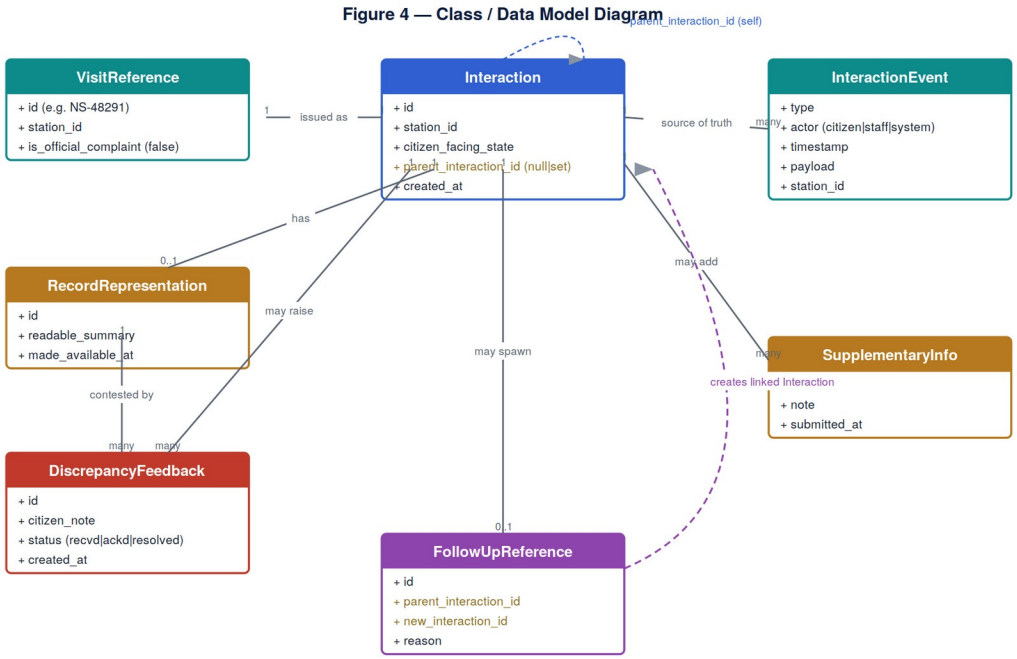


Figure 4 — Class / Data Model Diagram

The append-only InteractionEvent log is the source of truth; citizen- and station-facing views are projections. RecordRepresentation, the citizen's original description, and DiscrepancyFeedback are distinct entities so they never blur — the structural guarantee behind Fidelity. A FollowUpReference spawns a new Interaction (via new_interaction_id) that points back to its origin (via parent_interaction_id), forming a traceable chain of linked interactions rather than one interaction reopened repeatedly.

14. Out of Scope

NyaySetu improves visibility, understanding and continuity. It does not expand into adjacent problems, however tempting. The following are **explicitly not built** in the MVP — and a request that drifts toward any of them should be flagged and redirected to the in-scope path, not quietly accommodated.

- **Investigation.** No crime investigation, suspect identification, CCTV analysis, property recovery, or outcome determination.
- **Faster resolution.** NyaySetu makes no promise about speed of justice. Its promise concerns visibility, understanding and continuity — nothing more.
- **Police decision engine.** No determination of whether an offence occurred, which law applies, how something should be investigated, or who should investigate.
- **AI police officer.** No LLM or AI assistant in the MVP. The core problem is workflow visibility; adding intelligence before visibility solves the wrong problem.

- **Continuous location tracking.** No continuous GPS surveillance of citizens. Workflow is measured with events and timestamps only (see Section 15.4).
- **Full case-management system.** The Station Console does not manage investigation assignments, officer workload, evidence, suspect records, internal notes, or supervisor performance.
- **Misconduct / oversight adjudication.** The MVP captures a citizen's disagreement with a recorded representation. It does not run disciplinary investigations, corruption inquiries, complaint adjudication, or independent oversight workflows.
- **Nationwide integration.** No attempt to integrate every station, every state system, courts, or all government databases. One station, one journey, validated first.

15. Metrics

NyaySetu's metrics answer one question — did the core promise get delivered? — and they are instrumented from day one, never bolted on afterward. All analytics use privacy-conscious identifiers; no personally identifying information enters the analytics pipeline.

15.1 North Star — Verified Citizen Continuity Rate

The percentage of eligible interactions in which the citizen reaches all four of:

```
Visit created  +  Interaction acknowledged  +  Recorded information reviewed  +
Interaction Home / next-step visibility
```

This single measure captures whether Entry, Fidelity and Continuity actually landed for a real citizen — not whether a screen was built.

15.2 Supporting metrics

Grouped by the three pillars, plus the station workflow that makes them possible:

- **Entry** — Visit-creation completion rate; Visit-creation abandonment rate; time to create a Visit Reference.
- **Interaction** — time from Visit Reference creation to acknowledgement; percentage of interactions acknowledged.
- **Fidelity** — record-review rate; record-confirmation rate; discrepancy-flag rate; time from discrepancy submission to acknowledgement.
- **Continuity** — Interaction Home view rate; next-action comprehension; follow-up reference usage; supplementary-information submission rate.
- **Station workflow** — median time to acknowledge; time to publish a structured update; number of manual steps per interaction; percentage of interactions requiring duplicate data entry.

15.3 Guardrail metrics — do no harm

Visibility must never be improved at the cost of these. Each is tracked and alerted on:

- privacy complaints;
- incorrect citizen-facing status;
- false impressions of case registration;

- mismatched records;
- duplicate Visit References;
- increased frontline processing time;
- unauthorised access attempts;
- accessibility-related drop-offs.

15.4 Measuring without surveilling

Service quality is measured from **workflow timestamps**, not location:

```
Visit Reference created → Interaction acknowledged → Record created → Citizen
verification completed
```

This yields meaningful service metrics — how long a citizen waited to be acknowledged, how long until a record was available — with far less privacy risk than tracking a person. A geofence may be conceptually useful for convenience ("looks like you're near this station"), but it must never become continuous location tracking. The event log already answers the questions that matter.

15.5 A note on framing

Every metric above is a **proposed MVP success measure**, not a measured outcome. Because primary interviews were not conducted (Section 5), these are the instruments by which the hypothesis will be tested in usability validation and a controlled pilot — they are not claims about results already achieved.

16. Key Assumptions & Risks

16.1 Key assumptions

Because primary interviews and production workflow access were not available in the case-study timeline, the following remain assumptions to validate before production rollout:

6. Citizens value a traceable acknowledgement of their attempt to seek help.
7. Citizens can understand the distinction between a Visit Reference, recorded information, and an official complaint or case.
8. Record verification raises citizen confidence without materially slowing frontline work.
9. Meaningful updates and a clear next action reduce unnecessary follow-up visits.
10. Station workflows can expose enough citizen-safe information to create meaningful continuity.
11. A structured update model creates less frontline burden than free-form citizen communication.
12. A lightweight Station Console can support the MVP without becoming a duplicate police operating system.

16.2 Key risks and mitigations

Risk	Mitigation
NyaySetu creates false expectations of investigation or resolution	Explicitly distinguish Visit Reference, acknowledgement, recorded information, official complaint, and investigation — in copy and in the state model (Section 7.1).

Risk	Mitigation
Station adoption becomes the bottleneck	Minimise duplicate entry; use structured actions, not free-form updates; retrieve existing context rather than re-asking.
The Station Console drifts into a second police system	Hold strict scope: the console manages citizen-continuity states only (Section 8, H/I).
Citizens misread record confirmation as legal validation	Confirmation means "this reflects what I understand was recorded" — never legal validity or outcome.
Discrepancy feedback implies the record was corrected	Keep feedback received / feedback acknowledged / official-record change strictly separate (Section 7.2).
Sensitive operational information is exposed to citizens	Publish only approved citizen-safe states; enforce RBAC and privacy review; make arbitrary internal text architecturally unpublishable.
Updates become meaningless noise	Every state must answer "what happened?" and/or "what should I do next?" (Section 7.1).

17. Rollout Philosophy

NyaySetu does not launch nationally on day one. It earns each step.

```
Prototype
   ↓
Citizen usability testing  +  Station workflow testing
   ↓
Single-station controlled pilot
   ↓
Validate Entry → Fidelity → Continuity, and validate staff burden
   ↓
Limited regional rollout
   ↓
Interoperability expansion
```

The first production question is deliberately **not** "how do we integrate every police system in India?" It is:

> **"Does this interaction model meaningfully improve citizen clarity without breaking frontline operations?"**

Only once that is answered does the ingress boundary (Section 6.5) get pointed at real workflow events instead of the console, and only then does scale become the conversation.

18. Success Criteria / Definition of Done

The MVP is **directionally successful** when usability testing shows that most target citizens can complete the golden journey and correctly answer:

13. Was my interaction acknowledged?
14. What information was recorded?

15. What do I need to do next?

...and most station-side users can:

16. find a citizen interaction;
17. understand the submitted context;
18. acknowledge the interaction;
19. publish the correct structured citizen-facing state without confusion —

with **no material increase in frontline burden** and **no duplicate data entry.**

If either side cannot complete its core job, the interaction model is incomplete. That is treated as a **P0**, not a nice-to-have — because a citizen experience without a working station workflow behind it is exactly the "updates by magic" failure NyaySetu was built to avoid.

NyaySetu does not promise to fix policing. It does not promise to solve crime. It does not promise faster justice. It starts with something more fundamental:

> **A citizen should not need power, legal knowledge, or repeated physical visits just to know they were heard, understand what was recorded, and know what happens next.**

A meaningful citizen experience requires a meaningful workflow behind it. That is why NyaySetu is a Citizen Experience Layer and a lightweight Station Continuity Layer — together, the bridge between institutional workflow and citizen understanding.

19. References

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